HAZBOT / STUDY METHODOLOGY

Offline, source-bound analysis

01 SOURCES

Mission specification
7 phases · 12 Booleans

URDF chain
L1 = L2 = 0.100 m

Wheel gateway
SET: branch · 9600 baud

Arm gateway
5 aliases · 4 channels

02 ANALYSES

A Coordinator verification
$7 \times 2^{12} = 28{,}672$ pairs
vs. independent oracle · BFS
3 guard ablations

B Wrist geometry
URDF → analytic forward kinematics
$41^3 = 68{,}921$ joint samples
1,013 independent FK poses

C Command interfaces
261,121 wheel pairs (AST-isolated)
2,255 arm-mapping cases
5 aliases → 4 channels

03 EVIDENCE

Reference policy
0 violations (full)
480 · 128 · 288 ablated
109 checks pass

Geometry
0.200 m link-sum bound
2,747 samples below z = 0
FK Δ 6.8×10^{-17} m

Interfaces
1,020 mixed-zero: no send
magnitudes discarded
arm map 2,255/2,255

04 NEEDS

Handoff
explicit stop and
base-rest feedback

Docking
calibrated contact
workspace and pose

Gateway
keep wheel-rate
semantics

Provenance: SHA-256 source and data hashes · seeded sampling · 29 + 109 automated checks · stdlib Python only

Schematic